



Simple Molecules & Covalent Bonds

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The formation of covalent bonds

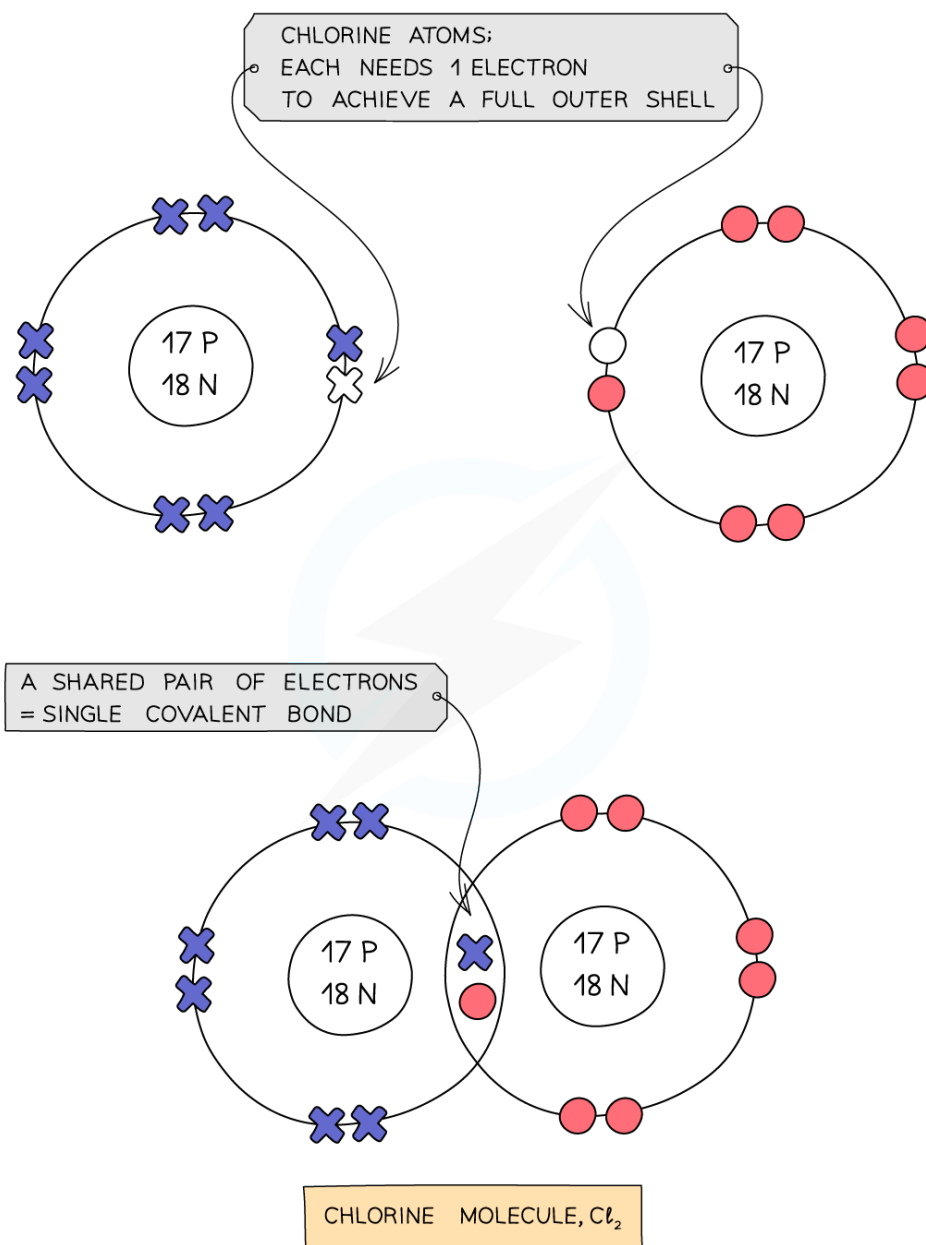
Covalent compounds

- Covalent compounds are formed when **pairs** of electrons are **shared** between atoms
- Only **non-metal elements** participate in covalent bonding
- As in ionic bonding, each atom gains a **full outer shell** of electrons, giving them a noble gas electronic configuration
- Covalently bonded substances may consist of small molecules or giant molecules
- When two or more atoms are **covalently** bonded together, we describe them as '**molecules**'
- **Dot-and-cross** diagrams can be used to show the electronic configurations in simple molecules
 - Electrons from one atom are represented by a dot, and the electrons of the other atom are represented by a cross
 - The electron shells of each atom in the molecule overlap and the shared electrons are shown in the area of overlap
 - The dot-and-cross diagram of the molecule shows clearly which atom each electron originated from

Diagram to show the formation of a covalent bond



Your notes



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Diagram showing how a covalent bond forms between two chlorine atoms



Examiner Tips and Tricks

When drawing dot-and-cross diagrams for covalent compounds, make sure that the electron shell for each atom is full (remember that the 1st shell can only hold 2 electrons).

Single covalent bonds

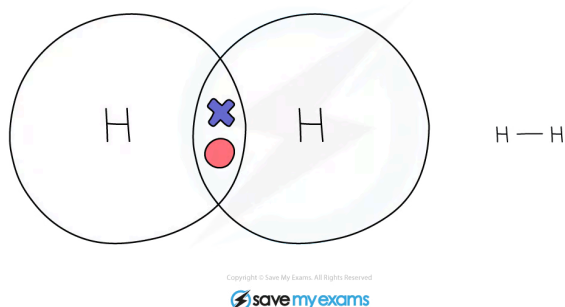
- Many simple molecules exist in which two adjacent atoms share one pair of electrons, also known as a single covalent bond (or single bond)



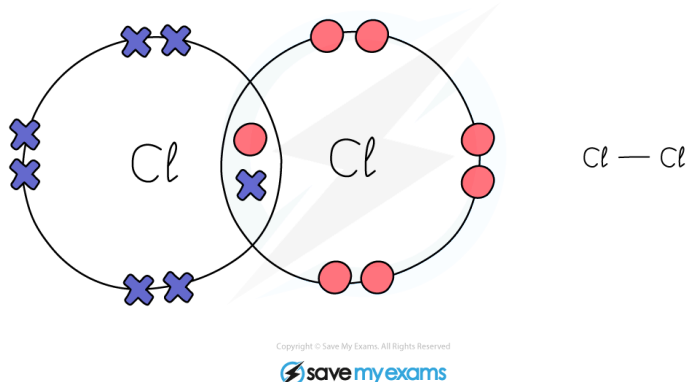
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Common Examples of Simple Molecules

Hydrogen:



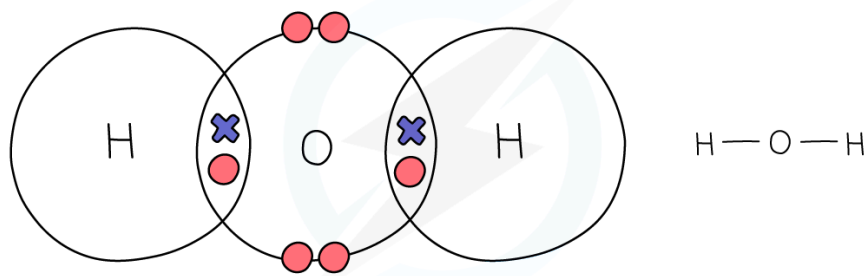
Chlorine:



Water:



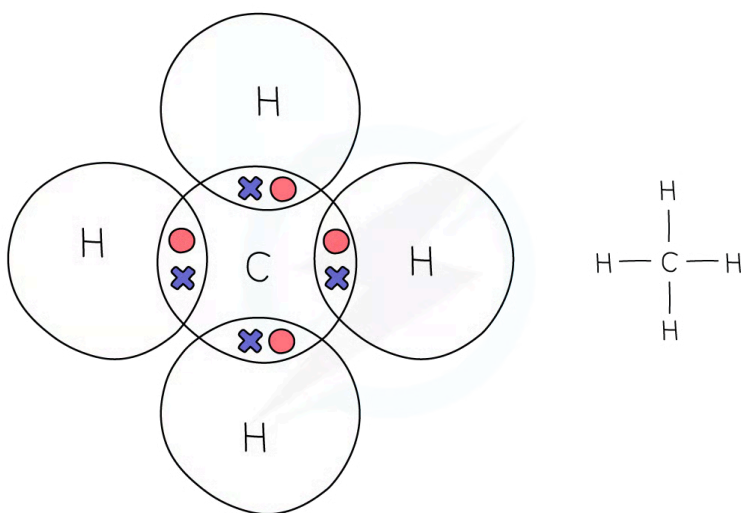
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Methane:



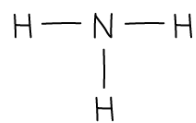
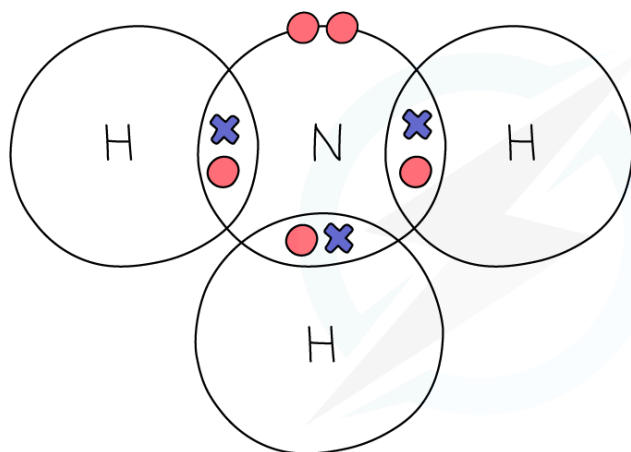
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Ammonia:



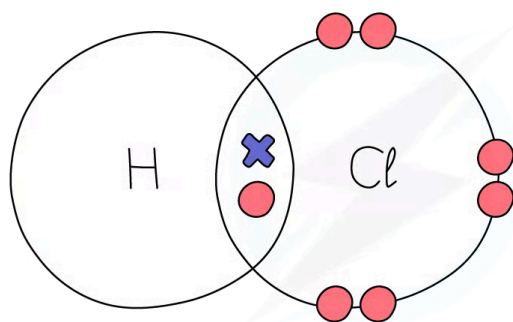
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Hydrogen chloride:



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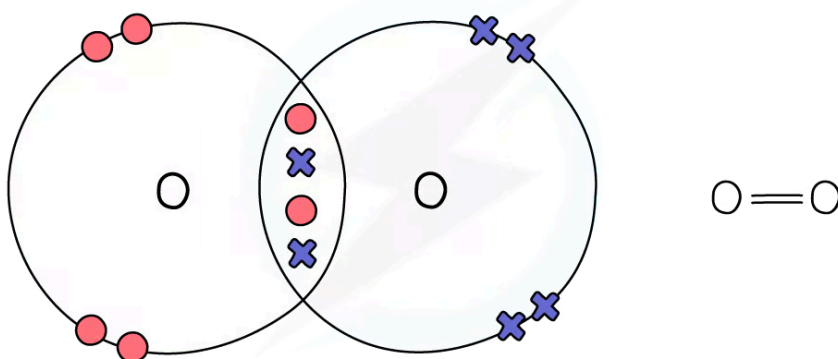
Covalent bonds in complex covalent molecules

Extended Tier Only

- Some atoms need to share more than one pair of electrons to gain a full outer shell of electrons
- If two adjacent atoms share two pairs of electrons, two covalent bonds are formed, also known as a **double bond**
- If two adjacent atoms share three pairs of electrons, three covalent bonds are formed, also known as a **triple bond**

Oxygen:

When 2 oxygen atoms react, they share 2 pairs of electrons to form a double bond



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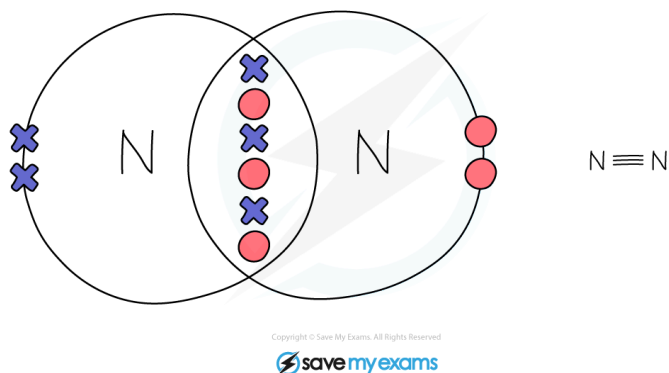


Nitrogen:

- When 2 nitrogen atoms react, they share 3 pairs of electrons to form a triple bond

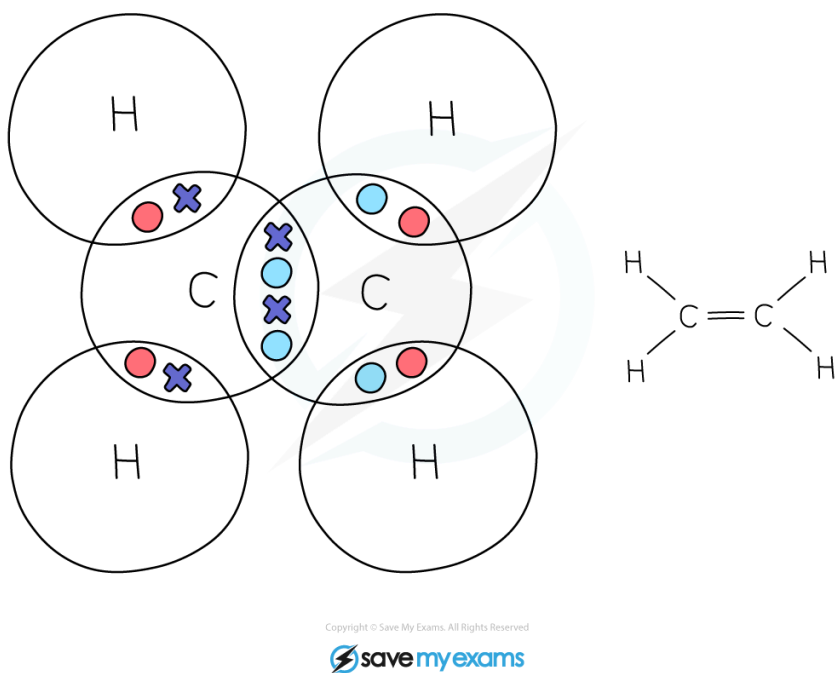


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Ethene:

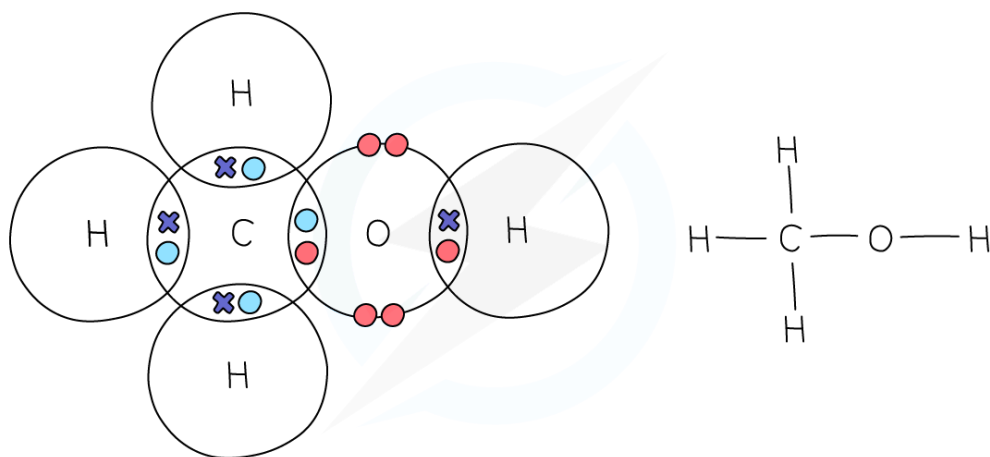
- In ethene, the 2 carbon atoms share 2 pairs of electrons
- This is known as a double bond



Methanol:



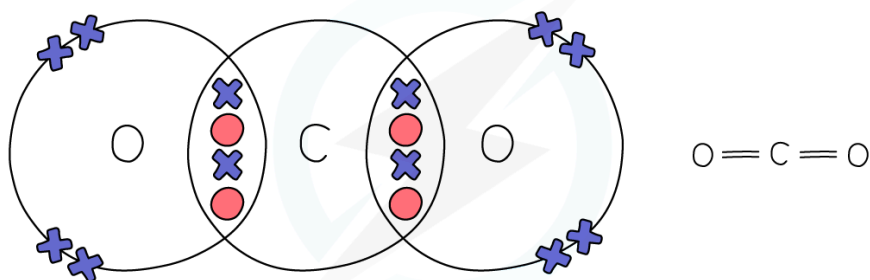
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Carbon Dioxide:



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Examiner Tips and Tricks

Be careful when drawing dot-and-cross diagrams, it is a common mistake for students to draw the wrong type of diagram.

Remember, if the compound contains **metal** and **non-metal**, it is an **ionic** compound and you need to draw the ions separated, with square brackets around each ion, together with a charge.

If the compound contains **non-metal** atoms only, it is a **covalent** compound, the shells should overlap and contain one or more pairs of electrons.



Properties of simple molecular compounds

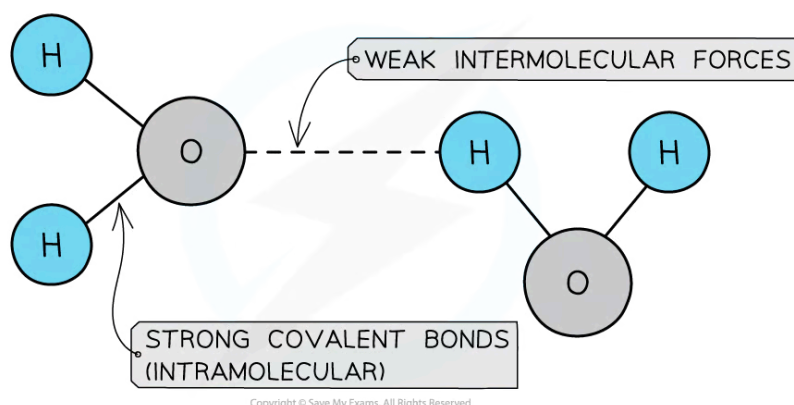
- Small molecules are compounds made up of molecules that contain just a few atoms **covalently bonded** together
- They have **low** melting and boiling points so covalent compounds are usually **liquids** or **gases** at room temperature
- As the molecules increase in **size**, the melting and boiling points generally increase
- Small molecules have poor electrical conductivity

Explaining the properties of simple molecular compounds

Extended tier only

Linking bonding and properties

- Small molecules have covalent bonds joining the atoms together, but weak intermolecular forces that act between neighbouring molecules
- They have **low** melting and boiling points as there are only **weak intermolecular** forces acting **between** the molecules
- These forces are **very weak** when compared to the covalent bonds and so most small molecules are either gases or liquids at room temperature
- As the molecules increase in **size** the intermolecular forces also increase as there are more electrons available
- This causes the melting and boiling points to **increase**



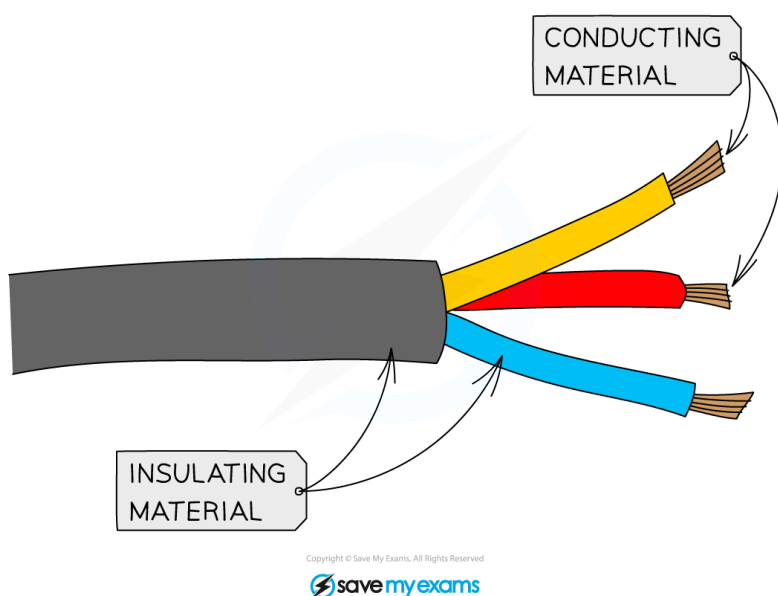
The bonds between hydrogen and oxygen in water are **COVALENT**, and the attractions between the molecules are **INTERMOLECULAR FORCES** which are about one tenth as strong as covalent bonds



Your notes

Electrical Conductivity

- Molecular compounds are **poor conductors** of electricity as there are no free ions or electrons to carry the charge.
- Most covalent compounds do not conduct at all in the solid state and are thus **insulators**
- Common insulators include the plastic coating around household electrical wiring, rubber and wood



The plastic coating around electrical wires is made from covalent molecules that do not allow a flow of charge



Examiner Tips and Tricks

The atoms within covalent molecules are held together by covalent bonds while the molecules in a covalent substance are attracted to each other by intermolecular forces.